

PISA

PISA 2025 Technical Report

Draft version



25 International Data Products

The international data products include public-use files (PUFs), codebooks and compendia tables. These products are available on the OECD website (<http://www.oecd.org/pisa/>). The IEA International Database Analyser (IDB Analyser, or IDB Analyzer) is configured for use with PISA data and can be downloaded from www.iea.nl.

Public-use files

The public-use files (PUFs) contain response records from all participating countries/economies included in the PISA sample. Student-level files contain responses to the background questionnaires and cognitive assessments, sampling weights, proficiency estimates, and variables derived from responses to the background questionnaires (BQs). The school and teacher files contain responses to the background questionnaires, variables derived from them, and school-level sampling information. The PUF variables represent a subset of those collected across all participating countries/economies and are available on the OECD website at <http://www.oecd.org/pisa/>.

Variables excluded, suppressed or recoded for some or all countries

The PUFs include only a subset of the variables included in the individual country data files.

Data collected through national adaptations and extensions are not included in the PUFs. In addition, selected variables were excluded or recoded following consultation with the OECD Secretariat because they: (i) have little or no analytical utility, (ii) were intended for internal or interim purposes only, (iii) relate to secure item material, or (iv) include personally identifiable data, or data that may increase the risk of unintended or indirect disclosure.

The following groups of variables are excluded from the PUFs:

- national adaptations and additional items in the background questionnaires
- direct, indirect, and operational identifiers for respondents
- open-ended text-entry responses to cognitive and background questionnaire items
- original scale score values (theta) before standardisation to an international metric

Countries/economies were also given the option to suppress selected variables in the PUFs. Suppression requests were approved when the data were judged to pose a risk to the anonymity of students, schools and/or teachers. Suppressed values are represented in the database using designated missing-value codes. For a selection of variables, recoding was applied for all countries/economies to minimise re-identification risks. In addition, some countries/economies requested specific recoding or suppressing of their own data. The list of recoding for all countries and economies is detailed in Chapter 13. The list of variables suppressed or recoded in the PUF in response to the requests from the participating countries and economies can be found in Annex Table 13.A.4.

Data files

Data files are provided in SPSS format. The files include:

- Student questionnaire data file: This file includes ID variables, response data, scaled and derived variables from the student, the Information, Communication and Technology (ICT) familiarity and parent questionnaires, plausible values for the core and innovative domains (Science, Mathematics, Reading and Learning in the Digital World [LDW]) and subscales (Science and LDW), and overall and replicate student weights.
- School questionnaire data file: This file includes ID variables, response data, scaled and derived variables from the school questionnaire, and an overall school weight.
- Teacher questionnaire data file: This file includes ID variables, response data, and scaled and derived variables from the teacher questionnaire.
- Core domains cognitive item data file: This file includes ID variables and raw and coded item responses for the core domains (Science, Reading and Mathematics).
- Core domains cognitive item process data file: This file includes computer-based assessment (CBA) cognitive item process data, such as total time and number of actions, for the core domains (Science, Reading and Mathematics).
- LDW cognitive item data file: This file includes ID variables, LDW raw and coded item responses, and item process data, such as total time and number of actions.
- LDW self-regulated learning data file: This file includes ID variables, indicators of self-regulation of behaviour, and associated student classifications.
- Questionnaire timing data file: This file includes computer-based assessment (CBA) questionnaire log data, such as total time on a unit or screen.

Additional data files containing data from the Foreign Language Assessment (FLA) are also made available:

- FLA cognitive item data file: This file includes ID variables, raw and coded item responses, and item process data for the FLA domains (FLA Listening, FLA Reading, FLA Speaking and PISA Reading).
- FLA student questionnaire data file: This file includes ID variables, response data, scaled and derived variables from the student questionnaire, plausible values for the domains assessed (FLA Listening, FLA Reading, FLA Speaking and PISA Reading), and overall and replicate student weights for the FLA sample.
- FLA questionnaire timing data file: This file includes questionnaire log data, such as total time on a unit or screen, for the FLA sample.

Variables used in sampling, weighting and merging

The variable STRATUM is included to identify sampling strata. The variable is created by concatenating a three-letter country code and a two-digit original stratum identifier.

The variables W_FSTUWT and W_FSTURWT1–W_FSTURWT80 represent the full student sampling weight and the 80 replicate weights used to estimate sampling variance.

The variable SENWT is a normalised weight variable typically used for analyses of student performance across a group of countries/economies where the contribution of each country/economy is intended to be equal, regardless of population or sample size. The senate weight sums to a constant of 5 000 across all cases within each country/economy in the file. This weight sums to 5 000 within each country/economy

only when there are no missing data for the variable of interest. The relative contribution of each country/economy is affected by the incidence of missing data.

The student and teacher data files can be merged with the school data file using the variable CNTSCHID. CNTSCHID combines the three-letter country/economy code and a randomised five-digit school ID number, making it unique across all countries/economies.

Data Quality Reviews and Updates

As part of the PISA 2025 data quality assurance process, the international databases were updated after completion of the analyses for the OECD PISA 2025 Results: Volume I. These updates corrected identified data issues and improved the accuracy and analytical utility of the data released in the Public Use File (PUF).

Accordingly, the PUF reflects these updates, while the analyses presented in Volume I remain based on an earlier database release, IDB2.1. All subsequent international publications and data products are based on the updated international databases incorporated in the PUF. The updates are summarised in Annex 25.A.

Codebooks for the PISA 2025 public-use data files

The PISA 2025 Main Survey data products include a set of data codebooks in Excel format. Each codebook is a printable report that provides descriptive information for each variable in the corresponding data file. The codebooks report frequencies and percentages for all categorical variables from the cognitive assessment and background questionnaires, as well as for variables derived and/or added during data processing. The codebooks are available from the OECD website (<https://www.oecd.org/pisa/data/>).

For each variable, the codebooks provide the variable name, variable label, values, and value labels. Other metadata are also provided, such as variable type (e.g., string or numeric) and precision/format. For non-categorical numeric variables, the codebooks additionally report the range of valid values (minimum and maximum). The codebooks for the main public-use data files are provided as separate worksheets within a single Excel workbook available from the OECD website. Each worksheet corresponds to one of the public-use data files described above.

Data compendia tables

Using the PUFs as the source data, the compendia comprise sets of summary tables that provide percentages for both cognitive and background items. The compendia support public-use file users by providing information about the contents of the data files and enabling users to confirm that their analyses of the PUFs are being performed correctly. The compendia are available on the OECD website (<http://www.oecd.org/pisa/>).

Questionnaire compendia provide the distribution of students across variables collected through the questionnaires. Cognitive compendia provide the distribution of student responses for each test item. Results are provided in Excel format, separately for background questions and test items, and are further disaggregated by questionnaire type and domain, and by gender for cognitive items. Each Excel file contains multiple worksheets, with each worksheet corresponding to a single variable. The first worksheet in each file is a table of contents that includes a hyperlink to each variable, allowing users to identify the available variables and navigate directly to the required data.

Separate tables provide percentage and percentile data for continuous background variables across all questionnaires.

All statistics included in the compendia are calculated from weighted data and are presented with their corresponding standard errors, which account for both sampling and measurement uncertainty. The OECD average is calculated as the simple average of the 38 OECD member countries as of PISA 2025.

Data analysis and software tools

Standard analytical packages for the social sciences and educational research do not readily recognise or support the complex PISA sample and assessment design. This limitation is addressed by the software tool made available to help database users access and analyse PISA data and produce basic outputs: the IEA's International Database Analyser (IDB Analyser). The IEA's IDB Analyser, used in conjunction with the PUFs, allows unit-record access to the public-use database and enables users to conduct analyses offline, derive additional variables, and produce estimates for further use and reporting. The IEA's IDB Analyser is described in the remainder of this chapter.

IEA's International Database Analyser

The IEA International Database Analyser (IDB Analyser) is an application developed by the International Association for the Evaluation of Educational Achievement (IEA) that can be used to analyse data from most major large-scale assessment surveys, including OECD studies such as PISA. Originally designed for international large-scale assessments, it can also be used with national assessments such as the United States National Assessment of Educational Progress (NAEP).

The IDB Analyser generates SPSS, SAS, or R syntax that can be used to analyse large-scale assessment databases. The syntax incorporates information from the sampling design when computing sampling variance and accounts for multiple plausible value imputations. The code generated by the IDB Analyser enables users to compute descriptive statistics and conduct statistical hypothesis testing across population groups without writing programming code.

The IDB Analyser is licensed free of charge, not sold, and is for use only in accordance with the terms of the licensing agreement. While anyone can use the software for free, users do not have ownership of the software itself or its components, including the SPSS, SAS, or R macros, and users are only authorised to use the SPSS, SAS, and R macros in combination with the IDB Analyser, unless explicitly authorised by the IEA. The software and licence expire at the end of each calendar year, when the user must download and reinstall the latest version of the software and agree to the new licence. A complete copy of the licensing agreement is included in the Appendix of the Help Manual of the IDB Analyser.

The analysis module of the IDB Analyser provides procedures for computing means, percentages, standard deviations, correlations, and regression coefficients for any variable of interest, both overall for a country and for specific subgroups within a country. It also computes percentages of people in the population who are within, at, or above performance benchmarks, user-defined cut points in the proficiency distribution, percentiles based on the achievement scale, or any other continuous variable.

The analysis module can be used to analyse PISA data files. It supports the following analyses:

- Percentages and means: Computes percentages, means, design effects, and standard deviations for selected variables by subgroups defined by the user. The percentage of missing responses is included in the output. It also computes t-test statistics for group mean differences, accounting for sample dependence.
- Percentages only: Computes percentages by subgroups defined by the user.
- Linear regression: Computes linear regression coefficients for selected variables that predict a dependent variable by subgroups defined by the user. The IDB Analyser can include plausible

values as dependent or independent variables in the linear regression equation. It can also contrast-code categorical variables (using dummy or effect coding) and include them in the linear regression equation.

- **Logistic regression:** Computes logistic regression coefficients for selected variables that predict a dichotomous dependent variable by subgroups defined by the user. The IDB Analyser can include plausible values as independent variables in the logistic regression equation. It can also contrast-code categorical variables and include them in the logistic regression equation. When used with SAS, users can also specify multinomial logistic regression models.
- **Benchmarks:** Computes the percentage of the population meeting a set of user-specified performance or achievement benchmarks by subgroups defined by the user. It computes these percentages in two modes: cumulative (percentage of the population at or above given points in the distribution) or discrete (percentage of the population within given points of the distribution). It can also compute the mean of an analysis variable for those at a particular achievement level when the discrete option is selected, as well as group mean and percentage differences between groups, taking sample dependence into account.
- **Correlations:** Computes correlations for selected variables by subgroups defined by the grouping variable(s). The IDB Analyser can compute correlations between sets of plausible values.
- **Percentiles:** Computes the score points that separate a given proportion of the distribution of scores by subgroups defined by the grouping variable(s).

When calculating these statistics, the IDB Analyser can use any continuous or categorical variable in the database, as well as scores in the form of plausible values. When plausible values are used, the IDB Analyser generates SPSS, SAS, or R code that accounts for the multiple-imputation methodology when calculating statistical variances, as applicable to the study.

All procedures offered within the analysis module of the IDB Analyser use appropriate sampling weights and standard errors for the statistics, computed according to the variance estimation procedure required by the design of the study.

Annex 25.A. Summary of International Database Updates

As part of the PISA 2025 data quality assurance process, the international databases were updated after completion of the analyses for the OECD PISA 2025 Results: Volume I (<https://doi.org/10.1787/73451bc5-en>). These updates corrected data issues and improved the accuracy and analytical utility of the data released in the PUFs.

ST059 variables (trend items) in the student questionnaire and FLA student questionnaire were removed due to translation issues identified across countries related to “How many <class periods> per week are you typically required to attend for the following subjects?” The following items have been dropped from the student background databases: ST059Q01TA, ST059Q02TA and ST059Q03TA, and the corresponding variables from the FLA databases: ST059FLQ01DA, ST059FLQ02DA, and ST059FLQ03DA.

Occupational Variables

An issue affecting the processing of ISCO occupation codes was identified in the preparation of the international databases affecting the transformation of occupation codes into ISEI¹ scores, the derivation of HISEI and, consequently, the computation of ESCS for 0.6% of the total assessed students.

Countries with more than 200 students with corrected values in the HISEI derived index were North Macedonia (affected n=899, 12.9% of total), Saudi Arabia (affected n = 671, 9.2% of total), Kurdistan Region (Iraq)* (affected n= 641, 11.2% of total) and Jordan (affected n = 373, 4.5% of total).

As HISEI is one of the three components used in the computation of the Economic, Social and Cultural Status (ESCS) index, the ESCS was recomputed for students with corrected HISEI values. Correlations between the original and revised ESCS exceed 0.99 in most countries/ economies with the exception of North Macedonia ($r = 0.98$) and Kurdistan Region (Iraq)* ($r = 0.98$). The relationship between the revised ESCS and science performance changes only negligibly after the correction. Therefore, the original Plausible Values (PVs) remained applicable, and no changes were made to the PVs.

Language of Questionnaire

Previous issues with the data in the variable (LANG QQQ) for Dushanbe (Tajikistan) and Israel* were resolved and data was added to the databases.

Note: See *PISA 2025 Notes on Participating Countries and Economies*.

¹ International Socio-Economic Index of occupational status.